

Lab Report 6: Implementing Convolutional Neural Networks (CNN) for Image Classification

Implementing a CNN on CIFAR-10

Course Code: COMP-341L

Course Name: Artificial Neural Networks Lab

Lab Number: 6

Lab Title: Implementing Convolutional Neural Networks (CNN) for Image Classification

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February 28, 2026

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1. Objective

Implement and evaluate a CNN on the CIFAR-10 dataset by completing all required tasks: dataset exploration, preprocessing, CNN design, training, evaluation, prediction, visualization, and analysis.

2. Task 1: Load and Explore the CIFAR-10 Dataset

- Displayed training and test dataset shapes: `x_train (50000, 32, 32, 3)`, `y_train (50000, 1)`, `x_test (10000, 32, 32, 3)`, `y_test (10000, 1)`.
- Visualized the 10 class names: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck.
- Displayed 10 sample CIFAR-10 images in a 2×5 grid.

3. Task 2: Preprocess the Data

- Normalized pixel values to the `[0, 1]` range by dividing by 255.
- Used sparse integer labels with `sparse_categorical_crossentropy` loss for multi-class classification.

4. Task 3: Build a CNN Model

The model architecture includes:

- Convolution layers with ReLU activation
- Max-pooling layers for downsampling
- Dropout for regularization
- Dense layers with ReLU
- Softmax output layer for 10-class classification

5. Task 4: Train the CNN (10 Epochs)

Metric	Training	Validation
Accuracy	0.7230	0.7488
Loss	0.7827	0.7341

6. Task 5: Evaluate the Model

6.1 Test Set Performance

Test accuracy: 0.7285 | Test loss: 0.7774

6.2 Confusion Matrix

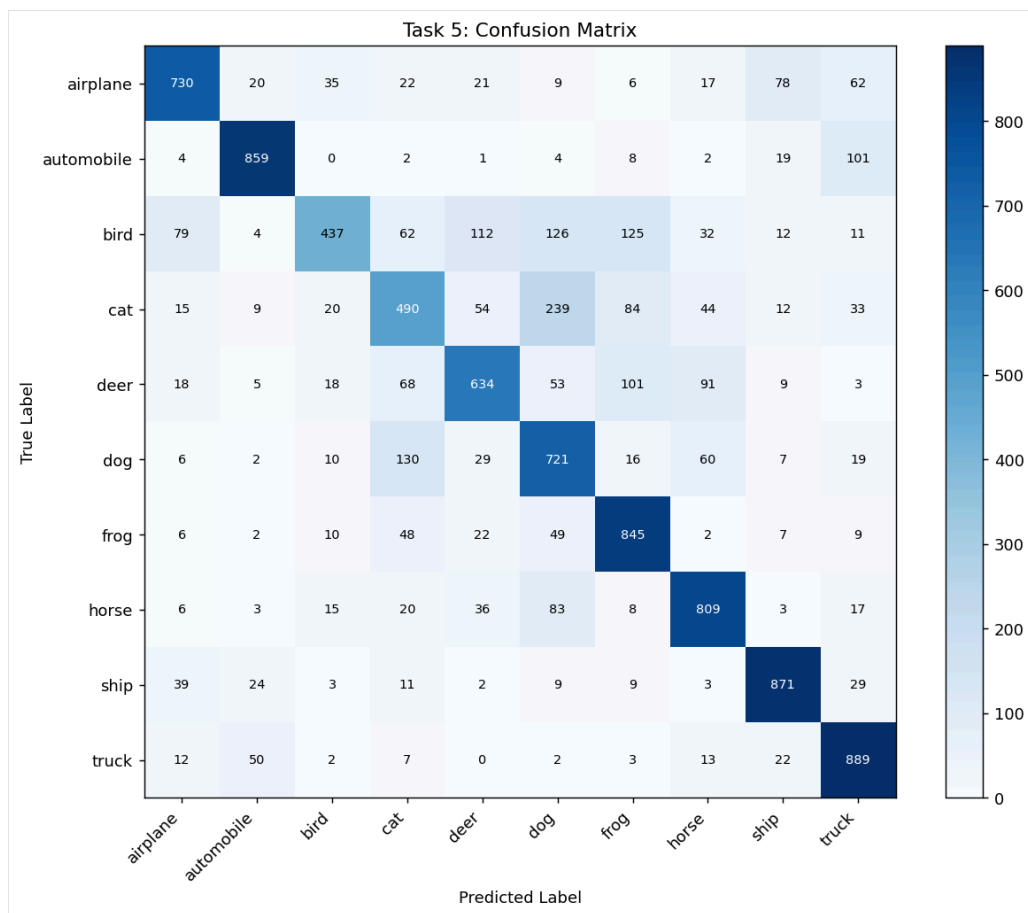


Figure 1. Confusion matrix showing predicted vs. actual labels on the CIFAR-10 test set.

6.3 Classification Report

Class	Precision	Recall	F1-Score	Support
airplane	0.7978	0.7300	0.7624	1000
automobile	0.8783	0.8590	0.8686	1000
bird	0.7945	0.4370	0.5639	1000
cat	0.5698	0.4900	0.5269	1000
deer	0.6959	0.6340	0.6635	1000
dog	0.5568	0.7210	0.6283	1000
frog	0.7012	0.8450	0.7664	1000
horse	0.7540	0.8090	0.7805	1000
ship	0.8375	0.8710	0.8539	1000
truck	0.7579	0.8890	0.8182	1000
accuracy			0.7285	10000
macro avg	0.7344	0.7285	0.7233	10000
weighted avg	0.7344	0.7285	0.7233	10000

7. Task 6: Make Predictions

Displayed 5 random test images with actual and predicted labels for qualitative inspection.



Figure 2. Five random test images with ground-truth and predicted labels.

8. Task 7: Plot Results

8.1 Training vs. Validation Accuracy

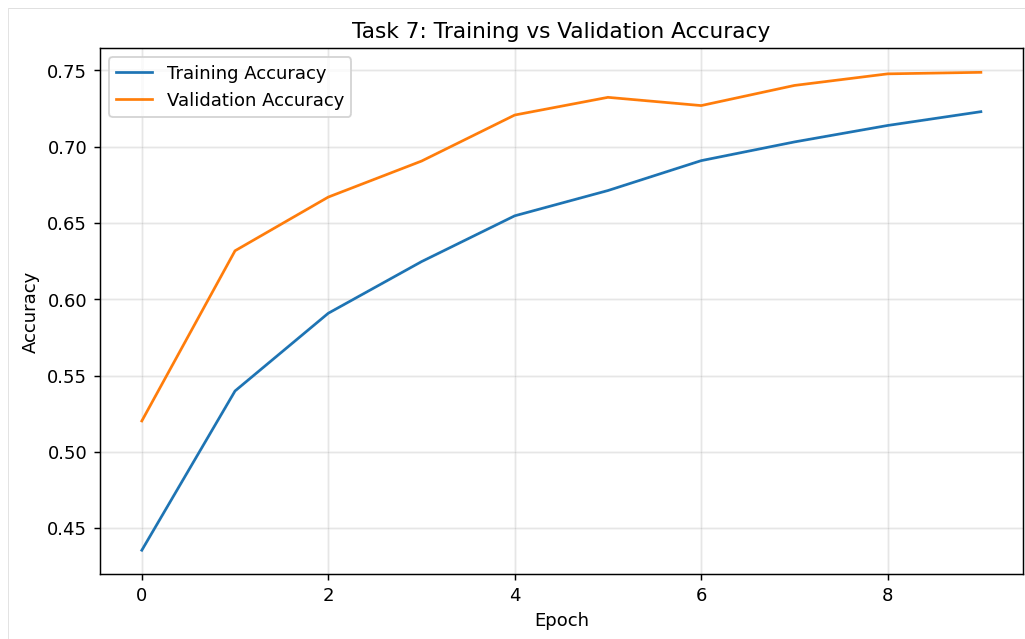


Figure 3. Accuracy curves over 10 epochs for training and validation sets.

8.2 Training vs. Validation Loss



Figure 4. Loss curves over 10 epochs for training and validation sets.

9. Task 8: Analysis

The CNN achieved good performance on CIFAR-10 with a test accuracy of **0.7285** after 10 epochs. Training accuracy was 0.7230 and validation accuracy was 0.7488 (gap = -0.0258), indicating limited overfitting. CIFAR-10 contains complex natural images, so moderate accuracy is expected with short training. Performance can improve with stronger data augmentation, deeper CNN blocks, better learning-rate scheduling, and longer training.

Per-class observations:

- **Best-performing classes:** automobile (0.87 precision), ship (0.84), truck (0.82)
- **Challenging classes:** cat (0.53 precision), dog (0.56), bird (0.44 recall)
- Cat–dog confusion is common due to visual similarity in low-resolution 32×32 images.

10. Conclusion

The required Lab 06 workflow was implemented successfully in Google Colab. The model was trained, evaluated, and analyzed using standard metrics and visual diagnostics. The CNN achieved approximately 73% test accuracy with minimal overfitting, and the confusion matrix and classification report provide clear insight into per-class performance.